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Engagement with alcohol marketing and early brand allegiance in relation to early years of drinking

En-Yi Lin, Sally Caswell, Ru Quan You, & Taisia Huckle

Social and Health Outcomes Research and Evaluation (SHORE), SHORE and Whariki Research Centre, Massey University, Auckland, New Zealand

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This study aimed to examine the relationship between measures of awareness to marketing and drinking among a sample of young people in New Zealand. The sample consisted of 1302 males and 1236 females predominantly aged between 13 and 14 years and drawn from a number of schools in a metropolitan city. They were surveyed using a computer assisted telephone interview.

Regression analyses examined relationships between marketing (awareness of and engagement with a range of alcohol marketing channels) and reports of brand allegiance and drinking status, drinking frequency and quantity and future drinking intentions.

The results showed that awareness of each alcohol marketing channel increased the odds of being a drinker by 8%. Engagement with web-based marketing increased the odds of being a drinker by 98% while engagement with traditional marketing increased the odds by 51%. Brand allegiance increased the odds of being a drinker by 356% and increased the likelihood of non-drinkers reporting future drinking intentions (by 73%). Brand allegiance was also associated with more frequent alcohol consumption (1.65 times more drinking occasions per year) and 86% more alcohol consumed on a typical occasion.

The results suggest that, while exposure to all forms of marketing are associated with drinking by young people, measures of more active engagement, such as owning merchandise and downloading screen-savers are stronger predictors of drinking. Having established a brand allegiance, at this early age, was related to not only drinking and future intentions to drink but also drinking patterns including consuming larger quantities.

Keywords: Alcohol, marketing, brand allegiance, adolescents, alcohol consumption

INTRODUCTION

Of all the areas which a public health approach to alcohol policy has to tackle, alcohol marketing is the one undergoing most rapid development and the area that has been most hotly contested in the research and policy arena (Casswell & Maxwell, 2005; Hastings, Anderson, Cooke, & Gordon, 2005; Jernigan & Mosher, 2005). Community concern over increases in heavy use of alcohol by younger people and research showing the impact of marketing on young people (Anderson, de Bruijn, Angus, Gordon, & Hastings, 2009; Babor et al., 2010) has meant that alcohol marketing is high on the political and public agenda internationally and in New Zealand.

In New Zealand a liberal policy which ensued after regulation of broadcast advertising was revoked in 1992 has allowed brand advertising to develop in all media with minimal constraints (Hill & Casswell, 2004). While traditional broadcast and print media achieve high reach among the general population, young people's media consumption is also increasingly inclusive of web-based vehicles. Interactive internet sites, on-line competitions, viral marketing and text-based campaigns reach a mobile population that actively tunes into and seeks out marketing campaigns. Word of mouth, utilised most effectively through viral campaigns, ensures quick uptake of new promotions and widespread reach at minimal cost (A. Kaitaki & J. Kaitaki, 2004). New alcohol beverage products, which provide a variety of potencies, tastes and prices

to meet the needs of young people at different stages of their drinking development, are also an important part of the alcohol marketing environment.

Establishing brand identity and brand allegiance has been recognized as an essential element of successful alcohol marketing (Casswell, 2004; Casswell & Zhang, 1998). Scholars have suggested that forming brand capital – that is, the accretion of meaning and positive affect associated with a particular brand – is one of the primary (and maybe the most important) functions of alcohol advertising (Casswell, 2004; Jernigan, 2009; McCreanor, Greenaway, Barnes, Borell, & Gregory, 2005; Novak, 2004; Saffer, 2002). A well-marketed brand is conceptualized as a personality with which users can form relationships and use as a critical resource for the construction of identity (Cova, 1996; Elliott & Wattanasuwan, 1998; Hanby, 1999; Fournier, 1998; Patterson & O'Malley, 2006; Warde, 1994). In New Zealand, longitudinal research identified the establishment of a relationship with a specific beer brand as an independent predictor of heavy drinking among young people: having a favourite alcohol (beer) brand was significantly associated with both current (at age 18) and future (at age 21) alcohol consumption (Casswell & Zhang, 1998). Other studies have also found that alcohol branding affects attitudes and drinking by young people (e.g. Austin, Chen, & Grube, 2006; Henriksen, Feighery, Schleicher, & Fortmann, 2008).

During the time period of increasing alcohol marketing in New Zealand (beginning in 1992) there have been measurable increases in consumption of alcohol by young people, particularly in terms of the quantities consumed resulting in higher annual volumes. Between 1995 and 2000, among males aged 14–15 years, the annual volume of alcohol consumed increased from four to nine litres, an increase to the equivalent of about 6 drinks per week. Among females aged 14–15, an increase from four to five litres of absolute alcohol annually was found (Habgood, Casswell, Pledger, & Bhatta, 2001).

The effect on the quantities of alcohol consumed by this age group is of interest. Previous research has found relationships between alcohol use in adolescence and later life (Donaldson, 2009; Monshouwer, Smit, De Zwart, Spruit, & Van Ameijden, 2003; Pitkanen, Lyyra, & Pulkkinen, 2005) and 'binge drinking' in particular has been found in longitudinal research to predict adverse social, educational and socioeconomic outcomes in later life (Viner & Taylor, 2007).

Most of the past research on alcohol marketing has focused on the more traditional channels for marketing, that is, advertising in the print and broadcast media (e.g. Snyder, Fleming-Milici, Slater, Sun, & Strizhakova, 2006) or alcohol branded merchandise items (e.g. Fisher, Williams Miles, Austin, Camargo, & Colditz, 2007) separately without investigating the cumulative impact of alcohol marketing across different channels. In a recent systematic review (Anderson

et al., 2009), only two out of the thirteen studies reviewed (i.e. Ellickson, Collins, Hambarsoomians, & McCaffrey, 2005; Collins, Ellickson, McCaffrey, & Hambarsoomians, 2007) looked into the impact of exposure to TV/magazine/radio advertising, in-store promotions, and concession stands (while Collins et al. (2007) also considered ownership of alcohol branded merchandise items) on adolescent alcohol use. To address this limitation, the current study goes beyond previous research by quantitatively examining a wider range of marketing techniques including new technologies such as websites, email and mobile phones, thus helps to demonstrate the extent, nature and reach of contemporary alcohol marketing.

The age range of the respondents for this study is 12–15 years with more than 90% falling between the ages of 13–14 years. In New Zealand, ages 14 and 15 years were the most common ages of first use of alcohol with the prevalence of alcohol use increasing from 15% at age 13 years to 61% by age 15 years (Huckle, Pledger, & Casswell, 2006). Therefore, this study aimed to investigate relationships between measures of alcohol marketing and drinking at this crucial formative time in relation to drinking; the age of onset.

The study design is similar to a recently published study in Scotland using a comparable questionnaire but different data collection (computer assisted telephone interviewing in New Zealand and face-to-face and self-completion in Scotland) (Gordon, Harris, Mackintosh, & Moodie, 2011). Gordon et al. (2011) found that the more aware and appreciative of alcohol marketing that young people were, the more likely they were to have consumed alcohol in the past 12 months, and the more likely non-drinkers were to say they would drink in the next 12 months. Our research hypotheses are that, in line with the Scottish finding, the more young people are aware of, and engaged with alcohol marketing, the more likely they are to have consumed alcohol (H1), and the more likely that non-drinkers will report intentions to drink alcohol in the next year (H2). We also hypothesize that young people who are more aware of, and engaged with alcohol marketing will drink more frequently and in greater amount than those who are less aware and responsive to alcohol marketing (H3). Our study also aims to examine relationships between brand allegiance and drinking behaviours. We hypothesized that those young people with a favourite alcohol brand will be more likely to be a drinker, more likely to intend to drink in the future and report higher alcohol consumption levels (H4).

METHODOLOGY

Design

This study is a cross-sectional analysis of the baseline data collected as part of longitudinal questionnaire-based design adapted from a tobacco marketing study in the UK (MacFadyen, Hastings, & MacKintosh, 2001). Questionnaire design and development was

based on findings from an earlier research phase and involved an extensive, iterative piloting process to ensure the acceptability, appropriateness and comprehension of questionnaire content for 12–15 years olds (Gordon et al., 2011). In the New Zealand study a question regarding brand allegiance which had been used in previous research (Casswell & Zhang, 1998) was added to the survey. The final survey consisted of an interview-administered questionnaire measuring awareness, engagement and involvement with alcohol marketing, brand allegiance and a set of questions measuring drinking and associated behaviours.

Procedure

All respondents were interviewed using a computer assisted telephone interviewing system (CATI) by a highly trained interviewing staff. Respondents were recruited via a) schools and b) random digit dialing. Most students (91%) were contacted via schools. The sampling procedure aimed to include a range of students and schools were stratified by region (rural and urban) and socio-economic status to include an appropriate mix of students. Every effort was made to recruit as many respondents as possible from the original pool of students and in excess of 90% of the students contacted via schools agreed to participate.

Where school principals gave written permission consent was not sought from parents or caregivers. The interviewing took place in school using cell phones to call back to the CATI system. In other schools all Year 9 students (first year of high school) were asked to take home an information sheet for parents/caregivers and a consent card along with a pre-paid return envelope. The students contacted in this way, along with those contacted by random digit dialing, were interviewed on their home phone also using the CATI system. For young people recruited by random digit dialing consent was obtained from a parent or guardian prior to the interview.

A total of 29 schools participated in the study. Schools were selected from the Auckland region stratified by area (rural and urban). We also ensured that there was a diverse range of schools from different socioeconomic areas (i.e. from decile 1 schools which are in the least deprived localities through to decile 10 schools which are in the most deprived localities). The vast majority of schools were secular. The sample included respondents of different ethnicity (Maori, Pacifica, Asian and Pakeha (of European or similar origin)).

The methodology was approved by the Massey University Human Ethics Committee. The total sample of our study consisted of 1302 males and 1236 females ($N = 2538$).

Measures

Awareness of alcohol marketing

Alcohol marketing awareness was assessed firstly with a single question: 'Can you think of any makes or

brands of alcohol that you have seen or heard advertised recently?' It was then followed by a set of questions assessing the awareness across 15 types of marketing channels (television/movies; large posters/billboards; in-store signs/posters; newspapers/magazines; merchandise items; special price offers; celebrity endorsement; unusual product design; sport sponsorship; music event sponsorship; television sponsorship; emails; websites; mobile/computer screen savers; social networking sites). Participants were asked if they had seen any alcohol marketing in each of the 15 channels listed (Yes/No/Don't Know) and if so what brands they could recall seeing marketed in this way. The total number of marketing channels reported was used in the following regression analysis to denote respondents' level of alcohol marketing awareness.

Engagement with alcohol marketing

To assess engagement with alcohol marketing, respondents were asked whether they had: received free samples of alcohol products, received free gifts showing alcohol brand logos, received special price offers for alcohol, received promotional mail or emails mentioning alcohol brands, owned clothing or other alcohol branded items (classified as traditional marketing), looked at web sites for alcohol brands, downloaded mobile phone or computer screensavers featuring alcohol brands and used social networking sites containing alcohol brands or logos (classified as web based marketing) (Yes/No/Don't know). Respondents who answered 'yes' to any of the five traditional marketing channels were classified as having engagement with traditional alcohol marketing. Respondents who answered 'yes' to any of the three web-based marketing channels were classified as having engagement with web-based alcohol marketing.

Brand allegiance

Alcohol brand allegiance was assessed with a single question: 'Do you have a favourite brand of alcohol?' (Yes/No).

Alcohol use

Drinking status was assessed with the question: 'In the last 12 months did you drink any beer, low alcohol beer or home brew/wine/spirits, liqueurs or mixed cocktails/sherry, port or vermouth/cider/ready-to-drink (RTD) drinks/other types of alcohol?' Those who responded 'yes' to any of the alcohol type were classified as drinkers.

Future drinking intention was assessed among non-drinkers via a single question: 'Do you think you will drink alcohol at any time during the next year?' with the following response categories: definitely not, probably not, probably yes, and definitely yes. Those who responded 'probably yes' or 'definitely yes' were classified as intended drinkers.

Frequency of alcohol consumed was assessed with the question: 'How often would you usually drink any

kind of alcohol?' Students' responses were converted into number of drinking occasions per year.

Amount of alcohol consumed was assessed with the question: 'On a typical occasion how much alcohol would you be drinking?' Respondents were asked to recall the type of containers and number of units they drank on a typical occasion for each alcohol type.

Control variables

Demographic control variables measured include age, gender, and ethnicity.

Alcohol-related control variables measured include parental attitudes towards drinking and alcohol consumption. These were assessed using the following measures: drinking behaviour ('Does you mother/father drink alcohol at all nowadays? Yes/No); perceived parental drinking approval ('Do you think your parents/caregivers would consider it ok or not ok for you to try drinking to see what it's like?' Ok/Not ok). Questions were also asked in respect to siblings drinking status and perceived approval ('Do you think your brother or sisters drink alcohol? Yes/No; 'Do you think your brother(s) or sister (s) would consider it ok or not ok for you to try drinking alcohol to see what it's like?' Ok/Not ok), and also friends' drinking status and perceived approval ('As far as you know, how many of your friends drink alcohol at least once a week? with response options ranging from all of them to none of them, those who had at least a few friends drink were grouped into one category; 'Do you think your closest friends would consider it ok or not ok for you to try drinking alcohol to see what it's like?' Ok/Not ok).

Analysis

Descriptive analyses were undertaken on measures of awareness and engagement comparing drinkers versus non-drinkers. Two by two tables were generated for each item and Chi square testing was undertaken to test for differences.

Two logistic regressions were performed with drinking status and drinking intention as the dependent variables. Independent variables were entered in the following sequence: (1) demographic control variables (age, gender, and ethnicity); (2) alcohol-related control variables (father/mother/sibling/peer's drinking behaviour and perceived drinking approval); and (3) alcohol marketing variables (awareness of alcohol marketing; engagement with alcohol marketing; and brand allegiance). For these analyses the total number of marketing channels an adolescent was aware of was utilised in the models as the measure of awareness. For the engagement measure items were divided into three groups a) engagements with traditional marketing only (Yes/No); b) engagements with web-based marketing only (Yes/No); and c) engagements with both traditional and web-based marketing. Alcohol marketing variables were entered one by one: Model 1 includes only awareness of alcohol marketing; Model 2 includes both awareness of alcohol marketing and engagement

with alcohol marketing; and Model 3 includes all three marketing variables.

Two general linear models were conducted with drinking frequency and drinking amount as the dependent variables. Independent variables were entered in the same sequence as the abovementioned logistic regression models. All statistical analyses were conducted using SAS 9.2. Results were deemed significant at the <0.05 level.

RESULTS

Drinking status, intentions and consumption

Of the total sample, 953 youths (38%) admitting drinking some alcohol in the last 12 months and 282 youths (currently abstainer) intended to drink alcohol next year (17.8%). The average geometric mean of annual drinking frequency was 7.6 (7.0–8.3) and the average amount on a typical drinking occasion was 29.7 ml (27.1–32.6).

Awareness of alcohol marketing

In relation to all marketing channels, except for TV sponsorship, drinkers showed significantly higher awareness of alcohol marketing compared to non-drinkers (Table I). About three quarters of these young drinkers were aware of alcohol marketing in TV, movies, billboards, marketing in liquor outlets and more than half were aware of merchandise and price offers. In regards to the digital media, one third of the drinkers reported awareness of screen savers and social networking sites (SNS). While lower percentages of non drinkers reported awareness of alcohol marketing the ranking of the media was similar.

Engagement with alcohol marketing

Table II shows young people's engagement with alcohol marketing. Significantly more drinkers reported engagement in alcohol marketing (all forms), whereas significantly more non-drinkers reported no engagement in any form of alcohol marketing.

Favourite alcohol brand

Of the total sample, 718 young people (28.3%) had a favourite alcohol brand, 71% of them were drinkers. T-test results suggested that young drinkers with a favourite alcohol brand drank significantly more frequently (11.2 *versus* 4.7, $p < 0.001$) and consumed significantly greater amount of alcohol on a typical occasion (49.2 mL *versus* 15.7 mL, $p < 0.001$). Young people who had a favourite alcohol brand were more likely to be a drinker than those without a favourite brand (71% *versus* 24%, $p < 0.001$). About a third of non-drinking young people who had a favourite brand (31%) intended to drink next year.

Logistic regression was performed to examine the associations between both the awareness of, and engagement with, alcohol marketing and reporting a favourite alcohol brand. Results suggested that, after

Table I. Youth drinkers and non-drinkers' awareness of alcohol marketing.

	Non-drinkers <i>N</i> = 1585		Drinkers <i>N</i> = 953		<i>P</i> value (χ^2 test for trend)
	%	<i>N</i>	%	<i>N</i>	
TV or movies	68	1059	72	682	0.018
Billboards	66	1041	76	715	<0.001
In-store signs or posters	62	978	73	696	<0.001
Newspapers or magazines	40	623	50	472	<0.001
Merchandise	39	618	60	571	<0.001
Price offers	42	651	55	523	<0.001
Celebrities	21	325	31	289	<0.001
Packaging	16	247	22	208	<0.001
Sport sponsorship	27	417	34	327	<0.001
Event sponsorship	13	205	21	200	<0.001
TV sponsorship	18	276	18	169	0.832
Emails	5	71	9	85	<0.001
Web sites	9	141	17	157	<0.001
Screen savers	16	257	36	337	<0.001
Social networking sites	17	273	37	352	<0.001

Table II. Type of promotional activity participated in, or used by, drinkers and non-drinkers.

Promotions	Non-drinkers <i>N</i> = 1585		Drinkers <i>N</i> = 953		<i>P</i> value (χ^2 test for trend)
	%	<i>N</i>	%	<i>N</i>	
Traditional marketing					
Received free samples	2	30	11	105	<0.001
Received free gifts	6	93	15	147	<0.001
Received special price offers	6	101	17	165	<0.001
Received promotional mail/email	7	118	14	132	<0.001
Owned merchandise items	15	240	34	321	<0.001
Web based marketing					
Looked at websites for alcohol brands	2	36	6	59	<0.001
Downloaded mobile phone or computer screensaver	3	46	9	89	<0.001
Used social networking sites containing an alcohol brands	5	79	18	175	<0.001
None of these	69	1100	42	400	<0.001

controlling for individual demographics and other people's drinking status and approval, awareness of each additional marketing channel increased the likelihood of having a favourite brand by 11% (OR = 1.11, 95% CI = 1.06 to 1.16), engagement in traditional marketing increased the likelihood of having a favourite brand by 63% (OR = 1.63, 95% CI = 1.28 to 2.08), and engagement in both the traditional and web-based marketing increased the likelihood of having a favourite brand by 148% (OR = 2.48, 95% CI = 1.78 to 3.45). Engagement in only the web-based marketing was not significantly associated with having a favourite brand (OR = 1.03, 95% CI = .62 to 1.70).

Alcohol marketing measures and drinking status, intention, and pattern

The first logistic regression model examines the association between drinking status and the number

of marketing channels adolescents were aware of and involvement in alcohol marketing, after controlling for the effects of various demographic and alcohol-related control variables (Table III). Our results suggested, after all the marketing variables were introduced, reported awareness of alcohol marketing in each channel was associated with increased probability of drinking of 8%. Young people who engaged with traditional alcohol marketing were 51% more likely to have drunk alcohol, while those who engaged with web-based alcohol marketing were 98% more likely to have drunk alcohol in the past 12 months. Young people engaged in both traditional and web-based marketing were 125% more like to have drunk alcohol. Brand allegiance appeared to be an important factor: having a favourite brand increased the odds of being a drinker by 354%. Family and friends drinking status and approval were also significantly related.

Table III. Association between alcohol marketing measures and self-reported drinking status (showing odds ratio and confidence intervals).

	Model 1	Model 2	Model 3
Age (12–13 years)			
14–15 years	1.24 (1.01–1.51)*	1.23 (1.00–1.51)*	1.23 (1.00–1.52)
Gender (male)			
Female	0.94 (0.77–1.13)	0.94 (0.77–1.14)	0.94 (0.77–1.15)
Ethnicity (European)			
Maori	1.25 (0.96–1.63)	1.18 (0.90–1.54)	0.80 (0.60–1.06)
Pacific	0.89 (0.67–1.18)	0.84 (0.63–1.12)	0.60 (0.45–0.83)**
Asian	0.70 (0.52–0.94)*	0.68 (0.50–0.91)*	0.70 (0.52–0.95)*
Drinking status			
Father	1.09 (0.88–1.36)	1.10 (0.88–1.37)	1.13 (0.90–1.42)
Mother	1.35 (1.09–1.67)**	1.36 (1.10–1.68)**	1.38 (1.10–1.73)**
Siblings	1.52 (1.23–1.87)***	1.48 (1.20–1.82)***	1.36 (1.09–1.70)**
Friends	1.98 (1.61–2.43)***	1.90 (1.54–2.34)***	1.67 (1.34–2.08)***
Perceived drinking approval			
Parents	2.61 (2.06–3.29)***	2.55 (2.02–3.23)***	2.36 (1.85–3.01)***
Siblings	2.02 (1.60–2.55)***	2.01 (1.59–2.54)***	1.92 (1.50–2.46)***
Friends	2.00 (1.63–2.46)***	1.99 (1.61–2.45)***	1.80 (1.45–2.45)***
Alcohol marketing variables			
Awareness of alcohol marketing channels	1.17 (1.12–1.22)***	1.10 (1.06–1.15)***	1.08 (1.03–1.13)***
Engagement with alcohol marketing			
Traditional only		1.66 (1.32–2.09)***	1.51 (1.19–1.93)***
Web-based only		1.85 (1.15–2.95)*	1.98 (1.22–3.24)**
Both traditional and web-based		2.81 (1.99–3.96)***	2.25 (1.57–3.22)***
Favourite brand			4.56 (3.62–5.76)***

Notes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Reference groups are given in brackets.

When non-drinkers' intentions to drink alcohol next year are examined, only one alcohol marketing variable – namely, having already determined on a favourite brand – was significant (Table IV). Specifically, young people who have a favourite alcohol brand were 1.73 times more likely to say they will drink alcohol next year. Our results also show that perceived parental/sibling/friends' approval to drink had a significant association with future drinking intentions: youth who think their parents/sibling(s)/friends would consider it OK for them to drink alcohol were 2.39, 2.19, and 1.98 times respectively more likely to say they will drink alcohol next year. Young people whose friends drank alcohol at least once a week was 2.97 times more likely to say they will drink alcohol next year.

General linear model was employed to examine the relationship between alcohol marketing and young peoples' drinking patterns (Table V). Model 1 suggested that awareness of marketing by itself was significantly associated with drinking frequency (increased it by 4%), however, when marketing engagement variables were also entered into regression model (Model 2) the association became

non-significant. This suggested that marketing engagement variables largely accounted for the effect of marketing awareness variable on drinking frequency. When all marketing variables were introduced (Model 3), engagement with both the traditional and web-based marketing increased the frequency of alcohol consumption by 34% and having a favourite alcohol brand increased the frequency of alcohol consumption by 65%. Three other variables were also significantly associated with drinking frequency, that is, friends' drinking status, parental drinking approval and friends' drinking approval: 64%, 24% and 20% respectively.

When looking at drinking amount (Table VI), similar patterns were found as that with drinking frequency. Awareness of marketing by itself was significantly associated with drinking amount (increased typical drinking amount by 4%), however, when marketing engagement variables were also entered into regression model (Model 2) the association became non-significant. This again suggested that marketing engagement variables largely accounted for the effect of marketing awareness variable on drinking behaviour. Model 2 further suggested that young people engaged in web-based marketing drank 36%

Table IV. Association between alcohol marketing measures and self-reported drinking intention (showing odds ratio and confidence intervals).

	Model 1	Model 2	Model 3
Age (12–13 years)			
14–15 years	1.03 (0.76–1.40)	1.03 (0.76–1.40)	1.03 (0.76–1.40)
Gender (male)			
Female	0.92 (0.69–1.23)	0.93 (0.69–1.24)	0.95 (0.71–1.27)
Ethnicity (European)			
Maori	0.93 (0.62–1.39)	0.93 (0.62–1.40)	0.87 (0.58–1.31)
Pacific	0.53 (0.34–0.82)**	0.53 (0.34–0.82)**	0.50 (0.32–0.78)**
Asian	0.73 (0.48–1.11)	0.73 (0.48–1.10)	0.74 (0.49–1.12)
Drinking status			
Father	1.27 (0.92–1.75)	1.26 (0.91–1.75)	1.28 (0.92–1.76)
Mother	1.19 (0.86–1.64)	1.19 (0.87–1.64)	1.19 (0.86–1.64)
Siblings	1.34 (0.98–1.84)	1.33 (0.97–1.83)	1.32 (0.96–1.82)
Friends	3.13 (2.28–4.28)***	3.11 (2.27–4.26)***	2.97 (2.16–4.09)***
Perceived drinking approval			
Parents	2.44 (1.70–3.50)***	2.43 (1.69–3.50)***	2.39 (1.66–3.44)***
Siblings	2.25 (1.57–3.23)***	2.26 (1.57–3.23)***	2.19 (1.52–3.15)***
Friends	1.98 (1.46–2.68)***	1.97 (1.46–2.68)***	1.98 (1.46–2.68)***
Alcohol marketing variables			
Awareness of alcohol marketing channels	0.97 (0.91–1.02)	0.96 (0.90–1.03)	0.96 (0.90–1.02)
Engagement with alcohol marketing			
Traditional only		1.07 (0.74–1.53)	1.04 (0.72–1.49)
Web-based only		0.97 (0.44–2.17)	1.00 (0.45–2.23)
Both traditional and web-based		1.15 (0.62–2.16)	1.07 (0.57–2.01)
Favourite brand			1.73 (1.18–2.53)**

Notes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Reference groups are given in brackets.

less alcohol on a typical occasion, but those who engaged in both traditional and web-based marketing consumed 31% more alcohol on a typical occasion. When brand allegiance (having a favourite brand) was also entered into the regression model (Model 3), associations between marketing engagement and drinking amount still exist but to a lesser extent. Having a favourite brand was the alcohol marketing variable that showed strongest association with drinking amount (young people who have a favourite alcohol brand drank 86% more alcohol on a typical occasion). Ethnicity, sibling's and friends' drinking status were also significantly associated with drinking amount. Maori youths and Pacific young people consumed 93% and 96% more alcohol on a typical occasion compared to European origin young people. Siblings and friends' drinking status increased young peoples' drinking amount by 43% and 39%.

DISCUSSION

This study examined the association between alcohol marketing and drinking among young people in New Zealand. Our results suggested, similar to Gordon et al.

(2011)'s findings, exposure to alcohol marketing (measured in terms of awareness and engagement) was significantly associated with being a drinker even after taking into account demographic and alcohol-related social and family variables. Similar to children in Scotland (Gordon et al., 2011), drinkers were more aware of and involved with (all but one form of) alcohol marketing than non-drinkers.

Our study revealed the extent to which young people in New Zealand are exposed to and engage with a range of alcohol marketing channels and once again this showed a very similar pattern to that reported in Scotland. Large proportions of children were aware of marketing in traditional media such as TV and billboards. More than half of the drinkers were aware of marketing in liquor outlets (much of which is prominent in supermarkets, retail outlet windows and on the street verge in New Zealand) and a third owned branded merchandise. Marketing of this sort has been shown in longitudinal research to have an impact on the drinking of young people (Snyder et al., 2006).

The fast expanding digital media, which have been less well investigated to date, have been shown in this study to have strong associations with drinking status

Table V. Association between alcohol marketing measures and self-reported drinking frequency (showing odds ratio and confidence intervals).

	Model 1	Model 2	Model 3
Age (12–13 years)			
14–15 years	0.93 (0.80–1.07)	0.92 (0.79–1.06)	0.92 (0.79–1.06)
Gender (male)			
Female	1.05 (0.91–1.22)	1.04 (0.90–1.20)	1.02 (0.89–1.18)
Ethnicity (European)			
Maori	1.35 (1.13–1.62)**	1.30 (1.09–1.57)**	1.10 (0.91–1.32)
Pacific	1.34 (1.08–1.66)**	1.31 (1.05–1.63)*	1.12 (0.90–1.40)
Asian	0.98 (0.77–1.25)	0.99 (0.77–1.25)	0.99 (0.78–1.25)
Drinking status			
Father	0.99 (0.84–1.17)	1.00 (0.84–1.18)	1.01 (0.85–1.19)
Mother	0.99 (0.84–1.17)	1.00 (0.85–1.18)	1.02 (0.87–1.20)
Siblings	1.21 (1.03–1.41)*	1.20 (1.03–1.41)*	1.13 (0.97–1.32)
Friends	1.73 (1.47–2.03)***	1.71 (1.46–2.01)***	1.64 (1.40–1.92)***
Perceived drinking approval			
Parents	1.29 (1.10–1.52)**	1.28 (1.09–1.49)**	1.24 (1.06–1.45)**
Siblings	1.10 (0.94–1.30)	1.08 (0.92–1.27)	1.09 (0.93–1.28)
Friends	1.26 (1.07–1.49)**	1.27 (1.08–1.50)**	1.20 (1.02–1.42)*
Alcohol marketing variables			
Awareness of alcohol marketing channels	1.04 (1.01–1.07)**	1.02 (0.99–1.05)	1.01 (0.98–1.04)
Engagement with alcohol marketing			
Traditional only		1.09 (0.91–1.29)	1.06 (0.89–1.25)
Web-based only		0.85 (0.61–1.20)	0.88 (0.63–1.23)
Both traditional and web-based		1.40 (1.13–1.75)**	1.34 (1.08–1.66)**
Favourite brand			1.65 (1.41–1.92)***

Notes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Reference groups are given in brackets.

of young people. Awareness and engagement with digital media was less widespread compared with traditional media, nevertheless a third of these young drinkers were aware of screensavers and SNS marketing alcohol, 9% had downloaded a computer or mobile phone screen saver and almost one in five had participated in SNS which contained marketing material related to an alcohol brand. From the present results it appears that engagement with such digital media is the marketing variable most strongly associated with being a drinker, similar to the influence of friends' drinking and parental approval of drinking.

Our study indicated that while awareness of marketing was associated with drinking patterns (alcohol marketing awareness was associated with a 4% increase in both drinking frequency and amount), these associations became non-significant after introducing marketing engagement variables. This suggested that once drinking was established, engagement with marketing was a more important factor in affecting drinking patterns than the awareness of marketing.

Having already established a favourite brand at this age emerged as a key marketing variable in this study.

It was strongly associated with likelihood of being a drinker (increased odds of drinking by 356%) and were significantly associated with patterns of drinking once drinking was established (increased annual frequency by 65% and typical amount by 86%), it was also strongly associated with future drinking intentions.

The results confirmed the importance of family and friends' approval and drinking status on young peoples' drinking. As was found in Scotland (Gordon et al., 2011), perceived parental/sibling/friends' drinking approval and mother/siblings/friends' drinking status were all associated with increased likelihood of being a drinker and the expectations of future drinking, (although they did not predict the quantities consumed).

Building a brand relationship is a key aim of alcohol marketing (Casswell, 2004; Gordon, Hastings, McDermott, & Evans, 2008; Jernigan, 2009; McCreanor et al., 2005; McCreanor, Barnes, Kaiwai, Borell, & Gregory, 2008). Industry sources refer to their marketing budgets as 'investment behind brands' (Hill, 2000). Brand allegiance at 18 years has previously been found in a New Zealand longitudinal study

Table VI. Association between alcohol marketing measures and self-reported drinking amount (showing odds ratio and confidence intervals).

	Model 1	Model 2	Model 3
Age (12–13 years)			
14–15 years	1.09 (0.92–1.28)	1.08 (0.92–1.27)	1.08 (0.92–1.27)
Gender (male)			
Female	0.97 (0.83–1.13)	0.96 (0.82–1.12)	0.93 (0.80–1.09)
Ethnicity (European)			
Maori	2.51 (2.06–3.07)***	2.42 (1.98–2.96)***	1.93 (1.58–2.38)***
Pacific	2.49 (1.95–3.17)***	2.42 (1.91–3.08)***	1.96 (1.54–2.50)***
Asian	0.97 (0.74–1.26)	0.96 (0.73–1.26)	0.96 (0.74–1.24)
Drinking status			
Father	0.98 (0.81–1.18)	0.97 (0.81–1.17)	0.97 (0.81–1.17)
Mother	0.87 (0.72–1.04)	0.87 (0.73–1.04)	0.89 (0.75–1.07)
Siblings	1.54 (1.29–1.83)***	1.54 (1.29–1.83)***	1.43 (1.20–1.70)***
Friends	1.48 (1.24–1.76)***	1.46 (1.22–1.75)***	1.39 (1.17–1.65)***
Perceived drinking approval			
Parents	1.16 (0.99–1.42)	1.17 (0.97–1.39)	1.13 (0.96–1.35)
Siblings	1.03 (0.87–1.24)	1.01 (0.84–1.21)	1.03 (0.87–1.23)
Friends	1.14 (0.95–1.37)	1.15 (0.96–1.39)	1.08 (0.90–1.29)
Alcohol marketing variables			
Awareness of alcohol marketing channels	1.04 (1.01–1.07)*	1.02 (1.00–1.06)	1.01 (0.98–1.05)
Engagement with alcohol marketing			
Traditional only		0.96 (0.79–1.16)	0.93 (0.77–1.13)
Web-based only		0.64 (0.44–0.94)*	0.67 (0.46–0.97)*
Both traditional and web-based		1.31 (1.03–1.66)*	1.25 (0.99–1.58)
Favourite brand			1.86 (1.57–2.21)***

Notes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Reference groups are given in brackets.

to predict alcohol consumption at age 21 (Casswell & Zhang, 1998) and having a favourite brand advertisement (described as a measure of high receptivity to marketing) was found in a longitudinal study of U.S. high school students to predict the likelihood of becoming a drinker. (Henriksen et al., 2008). This study adds important information regarding alcohol brand allegiance at this early age. The relationship shown in this study goes beyond onset of drinking and intentions to drink to the drinking quantities reported. It is this measure which has shown considerable change in New Zealand over the period of expanding alcohol marketing and, given the harms associated with heavy use of alcohol by adolescents this is an important relationship suggesting receptivity to marketing may influence heavier drinking.

The main limitation of this study is its cross-sectional design. Although our results show an association between awareness of, and exposure to, marketing, and drinking behaviour it cannot assess causality and reverse causality cannot be ruled out. Also, there may be intervening variables between awareness and engagement with marketing which are not measured in this study, thereby under estimating the relationship with drinking. The sample did not aim

to be representative of New Zealand school students and transfer of findings beyond the sample therefore requires caution. The design does allow for examination of the associations between marketing and drinking relevant variables and the results provide a strong replication of the findings of the Scottish study (Gordon et al., 2011), and much longitudinal research (Anderson et al., 2009) suggesting that there is a strong association between awareness of and engagement with alcohol marketing by school children and their drinking status and between measures of receptivity to marketing and their drinking pattern and future intentions to drink.

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